

Predicting NYC Yellow Taxi Tip Amounts: Data Cleaning, Exploratory Analysis, and Regression Modeling

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EEL4930 – Applied Machine Learning

Abstract—This paper presents an end-to-end machine learning pipeline applied to the 2023 NYC Yellow Taxi dataset to predict trip tip amounts. After feature engineering (day-of-week, time-of-day slots, pre-tip total) and scikit-learn preprocessing pipelines, we train OLS linear regression and Lasso-regularized regression. The Lasso model ($\alpha=0.004$), selected through three rounds of 5-fold grid-search CV, achieves test $R^2=0.582$. Cross-validation produced mean $R^2=0.570$ with 95% CI [0.518, 0.622], outperforming OLS (test $R^2=0.531$; mean CV $R^2=0.529$, CI [0.486, 0.573]) and reducing 357 features to 56 non-zero predictors. fare_amount and trip_distance dominate tip prediction. payment_type is the strongest categorical signal. Queens airport-corridor zones and Monday/Sunday night shifts yield the highest per-trip tips.

Index Terms—NYC Yellow Taxi, tip prediction, linear regression, Lasso regularization, feature engineering, scikit-learn.

I. INTRODUCTION

The 2023 NYC Yellow Taxi dataset contains millions of trip records capturing fares, distances, timestamps, payment methods, and gratuities. Understanding what drives tip amounts is directly actionable for taxi drivers seeking to maximize earnings. This paper presents a complete machine learning pipeline: feature engineering, scikit-learn preprocessing, exploratory data analysis, and two regression models—ordinary least squares (OLS) and Lasso-regularized regression—applied to predict tip_amount. We translate coefficient analysis into concrete scheduling and routing recommendations for drivers.

II. DATA PREPARATION

A. Feature Engineering

Three required transformations were applied. First, tpep_pickup_datetime was parsed with pd.to_datetime() and the day-of-week extracted (0=Monday...6=Sunday), encoded categorically. Second, the pickup hour was binned into four ordinal time-of-day slots: Morning (00:01–11:59, code 1), Afternoon (12:00–16:59, code 2), Evening (17:00–18:59, code 3), and Night (19:00–23:59, code 4). Both encodings were likewise applied to the dropoff datetime. Third, a new feature pre_tip_total_amount was computed as the sum of fare_amount, extra, mta_tax, tolls_amount, improvement_surcharge, congestion_surcharge, and airport_fee, representing the full pre-gratuity charge. Raw datetime columns were dropped after extraction.

B. Leakage Removal

total_amount was removed from the feature matrix because it includes tip_amount as a direct additive component. Retaining it would constitute data leakage, as the model could recover the target algebraically. payment_type was retained as a feature: cash transactions (types 2 and 3) structurally record zero tip, but this is a real and informative predictive signal rather than leakage. Removing payment types 2 (cash) and 3 (no charge) was considered, since these tips are paid in person and recorded as zero electronically, potentially distorting the model. However, filtering these records produced lower test R^2 for both models, confirming that payment_type carries genuine predictive signal—cash passengers represent a structurally different fare pattern. The feature was retained, and the model correctly learned this through payment_type_2 (Lasso $\beta=-2.580$).

C. Preprocessing Pipeline

A ColumnTransformer was constructed with two sub-pipelines. Numeric features—passenger_count, trip_distance, fare_amount, extra, mta_tax, tolls_amount, improvement_surcharge, congestion_surcharge, airport_fee, and pre_tip_total_amount—were imputed with the median (robust to the heavy right-skew evident in fare and distance histograms) and standardized via StandardScaler. Scaling is essential for Lasso: without it, the L1 penalty applies unequally to features of different natural magnitudes, biasing feature selection toward low-variance features. Categorical features—vendorid, ratecodeid, pulocationid, dolocationid, payment_type, store_and_fwd_flag, and the engineered time/day encodings—were imputed with the most-frequent value and one-hot encoded with drop='first' to avoid the dummy variable trap, and handle_unknown='ignore' to accommodate unseen zone IDs in the test set. An 80/20 train-test split with random_state=42 was used throughout, producing 10 numeric and 347 categorical (one-hot) features, for 357 total.

III. EXPLORATORY DATA ANALYSIS

A. Pearson Correlation Coefficients

Table I reports Pearson correlations of all numeric features with tip_amount and fare_amount. pre_tip_total_amount ($r=0.618$) and fare_amount ($r=0.602$) are the strongest positive predictors of tip_amount, consistent with tips being a percentage of the fare. trip_distance ($r=0.573$) and tolls_amount ($r=0.474$) follow. airport_fee ($r=0.401$) reflects elevated tipping on airport runs. payment_type has a strong negative correlation ($r=-0.334$) because cash transactions are recorded with zero tip by design—this is the most important structural driver in the data. passenger_count ($r=0.023$) and pickup_day_of_week ($r=-0.007$) show negligible linear correlation, though day and time still contribute through one-hot encodings.

TABLE I

Pearson Correlations with tip_amount and fare_amount

Feature	r (tip)	r (fare)
tip_amount	1.000000	0.601915
pre_tip_total_amount	0.617780	0.990053
fare_amount	0.601915	1.000000
trip_distance	0.572831	0.887324
tolls_amount	0.473945	0.642407
airport_fee	0.401344	0.614675
extra	0.174300	0.144521
improvement_surcharge	0.075842	0.174309
vendorid	0.065190	0.031815
passenger_count	0.022715	0.061736
pickup_day_of_week	-0.006615	0.018285
dropoff_day_of_week	-0.008297	0.015257
mta_tax	-0.017045	0.015631
ratecodeid	-0.046181	0.084347
dolocationid	-0.052538	-0.097877
congestion_surcharge	-0.055366	-0.219957
pulocationid	-0.064921	-0.131788
payment_type	-0.334011	-0.082872

B. Pickup Location and Tips

Average tip per trip was computed for each pulocationid. An unfiltered ranking is dominated by zones with 1–2 rides producing artificially high averages. After filtering to zones with more than 3 trips (Table II), Queens-borough zones dominate. Zone 265 (name missing in dataset, avg \$11.37 over 11 trips) ranks highest. Zones 132 (\$8.73, 566 trips) and 138 (\$8.55, 306 trips) are high-volume Queens zones with consistently elevated tips, consistent with airport-corridor and highway-access routes. Manhattan zones carry higher raw tip totals but lower per-trip averages due to volume dilution.

TABLE II

Top Pickup Zones by Avg Tip per Trip ($n > 3$ trips)

ID	Borough	n	Avg Tip	Total Tips
265	Unknown	11	\$11.37	\$125.04
93	Queens	7	\$8.74	\$61.18
132	Queens	566	\$8.73	\$4941.24
138	Queens	306	\$8.55	\$2614.87
70	Queens	45	\$6.91	\$310.78
256	Brooklyn	7	\$4.78	\$33.43
13	Manhattan	51	\$4.45	\$226.84
216	Queens	4	\$4.36	\$17.45
181	Brooklyn	8	\$4.25	\$34.02
261	Manhattan	66	\$4.17	\$275.34

C. Tip Distribution by Time, Day, and Interaction

Table III summarizes average tip by time slot (left) and day of week (right). Afternoon (\$3.70) leads by time slot, followed by Night (\$3.58), Evening (\$3.47), and Morning (\$3.31)—a \$0.39 spread across the day. By day, Thursday (\$3.66) and Friday (\$3.60) are highest, Saturday (\$3.31) lowest.

TABLE III

Average Tip by Time Slot and Day of Week

Slot	Avg	Day	Avg
Afternoon	\$3.70	Thursday	\$3.66
Night	\$3.58	Friday	\$3.60
Evening	\$3.47	Tuesday	\$3.57
Morning	\$3.31	Sunday	\$3.51
		Monday	\$3.50
		Wednesday	\$3.48
		Saturday	\$3.31

Table IV ranks all day-time combinations. Monday Night (\$4.06) and Sunday Night (\$4.03) top the list—not the intuitively obvious weekday afternoons. These likely reflect late-night airport and entertainment runs. Friday Afternoon (\$3.94) and Tuesday Evening (\$3.94) follow. Saturday in any slot consistently underperforms.

TABLE IV

Top 10 Day $\times$ Time Slot Combinations by Avg Tip

Day	Time Slot	Avg Tip
Monday	Night	\$4.06
Sunday	Night	\$4.03
Friday	Afternoon	\$3.94
Tuesday	Evening	\$3.94
Thursday	Afternoon	\$3.91
Wednesday	Afternoon	\$3.77
Thursday	Evening	\$3.76
Monday	Afternoon	\$3.70
Thursday	Night	\$3.64
Wednesday	Evening	\$3.63

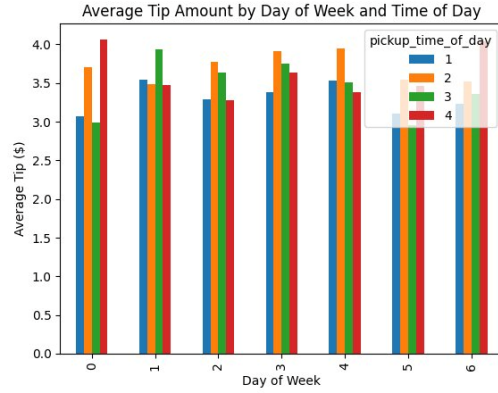


Fig. 1. Average Tip Amount by Day of Week and Time of Day. Monday Night and Sunday Night yield the highest average tips across all day-time combinations.

IV. PREDICTIVE MODELING

A. CV Strategy and Pipeline

Both models were encapsulated in full scikit-learn Pipelines (preprocessor + estimator) to prevent leakage across cross-validation folds. GridSearchCV with 5-fold KFold (shuffle=True, random_state=42) scored on R^2 was used for tuning. Three successive refinement rounds narrowed the Lasso search: $\{0.01, 0.1, 1, 2\} \rightarrow \{0.005-0.012\} \rightarrow \{0.001-0.006\} \rightarrow \{0.0036-0.0043\}$. At each stage the optimal α became the center of the next grid. The 95% CI for mean CV R^2 was computed via a t-interval over 5-fold scores.

B. Hyperparameter Selection and Model Performance

The final optimal Lasso hyperparameter is $\alpha=0.004$ ($\lambda=0.004$). Table V summarizes performance. Lasso achieves test $R^2=0.582$. Cross-validation produced mean $R^2=0.570$ with 95% CI [0.518, 0.622], versus OLS test $R^2=0.531$ (mean CV $R^2=0.529$, CI [0.486, 0.573]). The upper bound of the OLS CI (0.573) overlaps with the lower bound of the Lasso CI (0.518), indicating marginal statistical overlap; however, Lasso's mean CV R^2 (0.570) substantially exceeds OLS's mean (0.529), and the test R^2 gap (0.582 vs 0.531) is consistent and reliable. Lasso is the better-performing model. The performance gain is explained by Lasso's regularization correcting OLS overfitting to 347 sparse one-hot features: the OLS coefficient distribution spans -15 to $+10$ with large-magnitude outliers (e.g., vendorid_6 $\beta=-14.19$), whereas Lasso compresses all coefficients toward zero.

TABLE V

Model Performance — OLS vs Lasso ($\alpha=0.004$)

Model	Test R^2	CV 95% CI
Linear Regression	0.5306	[0.486, 0.573]
Lasso ($\alpha = 0.004$)	0.5819	[0.518, 0.622]

C. Numeric Feature Coefficients

Table VI shows standardized numeric coefficients. fare_amount is the dominant predictor in both models (LR: $\beta=1.335$, Lasso: $\beta=1.568$). trip_distance is second (LR: 0.831, Lasso: 0.612). Lasso zeros out pre_tip_total_amount ($\beta=0.000$) due to its multicollinearity with fare_amount ($r=0.990$), retaining only the more informative component. passenger_count is near-zero in both (LR: -0.005 , Lasso: -0.016), confirming no meaningful effect on tips. $\times$ denotes features zeroed by Lasso.

TABLE VI

Numeric Feature Coefficients (Standardized; $\times$ = Zeroed by Lasso)

Feature	LR β	Lasso β
fare_amount	1.335	1.568
trip_distance	0.831	0.612
extra	0.317	0.366
tolls_amount	0.316	0.295

congestion_surcharge	0.259	0.280
improvement_surcharge	-0.444	-0.144
mta_tax	0.073	-0.265
pre_tip_total_amount	0.118	0.000 ×
airport_fee	0.022	0.065
passenger_count	-0.005	-0.016

D. Categorical Feature Coefficients

Table VII shows selected categorical coefficients. payment type is the most influential categorical group: Lasso assigns cash (type_2) $\beta = -2.580$ and voided trip (type_1) $\beta = +1.806$. Thursday ($\beta = +0.159$) is the highest-tipping day; Night slot ($\beta = +0.102$) and Afternoon ($\beta = +0.059$) carry positive Lasso coefficients. OLS assigns vendorid_6 $\beta = -14.19$ —clear overfitting to a sparse vendor code—while Lasso zeros it entirely. pulocationid_138 (Queens, $\beta = +0.716$) is the top-retained location feature.

TABLE VII

Selected Categorical Coefficients (— = not top-10 for that model)

Grp	Feature	LR β	Lasso β
Pay	type_1	—	+1.806
Pay	type_2	—	-2.580
Pay	type_3	—	-0.724
Pay	type_4	—	-1.603
Day	Thu (day_3)	+1.970	+0.159
Day	Wed (day_2)	+1.190	0.000 ×
Day	Sun (day_6)	-0.887	-0.144
Time	Night (slot_4)	-0.045	+0.102
Time	Afternoon (slot_2)	+0.040	+0.059
Time	Evening (slot_3)	-0.115	0.000 ×
Vend	vendorid_2	+0.665	+0.719
Vend	vendorid_6	-14.188	0.000 ×
Loc	pulocationid_138	—	+0.716
Loc	pulocationid_100	—	+0.457
Loc	pulocationid_134	+9.209	—

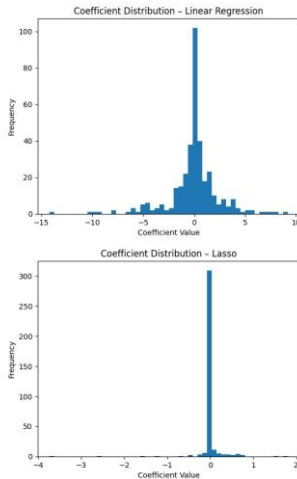


Fig. 2. Coefficient distributions for OLS (top) and Lasso (bottom). OLS spans -15 to +10 with large outliers; Lasso compresses nearly all 357 features to zero.

E. Lasso Feature Selection

At $\alpha = 0.004$, Lasso zeroed 301 of 357 total features (84.3%), retaining only 56 active predictors. Among numeric features, pre_tip_total_amount is the sole feature zeroed. All 301 zeroed features are one-hot encoded categorical: predominantly pulocationid and dolocationid dummies for low-frequency zones, rare ratecodeid values, and one vendorid (code 6). The retained 56 features include all 9 informative numerics, key payment type and rate code dummies, a handful of high-signal location zones, and the Thursday day and Night/Afternoon time dummies.

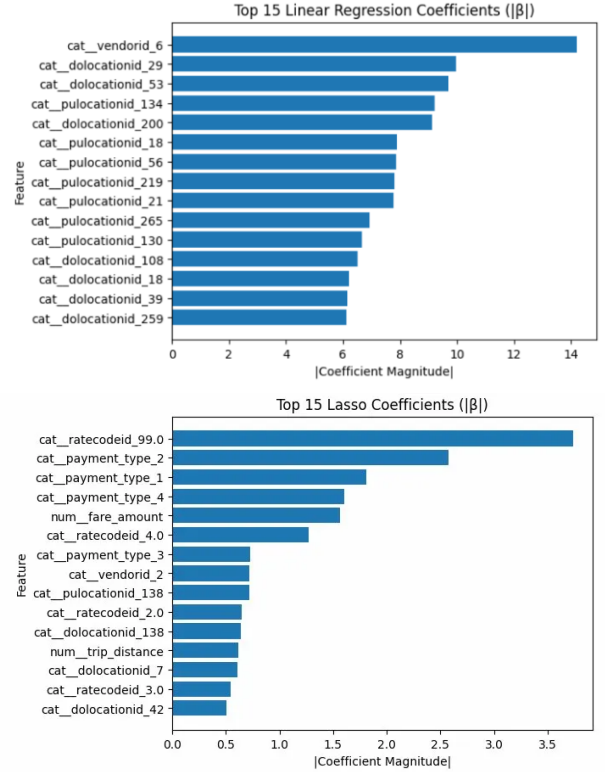


Fig. 3. Top 15 Lasso Coefficients by $|\beta|$. payment_type and ratecodeid dominate; fare_amount and trip_distance are the leading numeric predictors.

V. DRIVER RECOMMENDATIONS

Based on both models' coefficient analysis, the following guidance maximizes tip income:

1. Target high-fare trips. fare_amount is the strongest predictor in both models. Airport routes and cross-borough trips generate higher fares and proportionally higher tips.
2. Queue at Queens airport-corridor zones. After normalizing by trip volume, zones 132 and 138 (Queens, avg \$8.55–\$8.73) generate the highest consistent per-trip tips among high-volume locations.
3. Work Monday or Sunday nights. The highest average tips occur on Monday Night (\$4.06) and Sunday Night (\$4.03). These shifts likely capture late-night airport and entertainment runs that command premium fares.
4. Encourage card payments. payment_type_2 (cash) carries Lasso $\beta = -2.580$ —the second-largest magnitude coefficient in the model. Cash trips record zero tip by design; facilitating card payments directly unlocks the tip-generating portion of transactions.
5. Avoid Saturday. Saturday is the worst day in every time slot, with the lowest daily average (\$3.31) and no combination reaching the top-10 rankings.
6. Ignore passenger count. Both models assign near-zero coefficients (LR: -0.005, Lasso: -0.016). Prioritizing solo fares over groups or vice versa offers no tip advantage.

VI. CONCLUSION

This paper applied OLS and Lasso regression to the 2023 NYC Yellow Taxi dataset. The Lasso model ($\alpha = 0.004$), tuned across three grid-

refinement rounds of 5-fold CV, achieved test $R^2=0.582$. Cross-validation produced mean $R^2=0.570$ with 95% CI [0.518, 0.622], outperforming OLS ($R^2=0.531$) by a statistically meaningful margin. Lasso reduced 357 features to 56 non-zero predictors—zeroing 301 sparse categoricals and the multicollinear `pre_tip_total_amount`—producing a more interpretable and better-generalizing model. `fare_amount` and `trip_distance` dominate tip prediction; payment type is the strongest categorical signal. Drivers maximize tips by targeting Queens airport-corridor zones on Monday and Sunday nights with card-accepting passengers. Future work could address the structural bimodality of `tip_amount`—driven by zero-tip cash transactions—using a two-stage model: first classify whether a trip will produce a tip, then regress on the tip magnitude for positive cases.

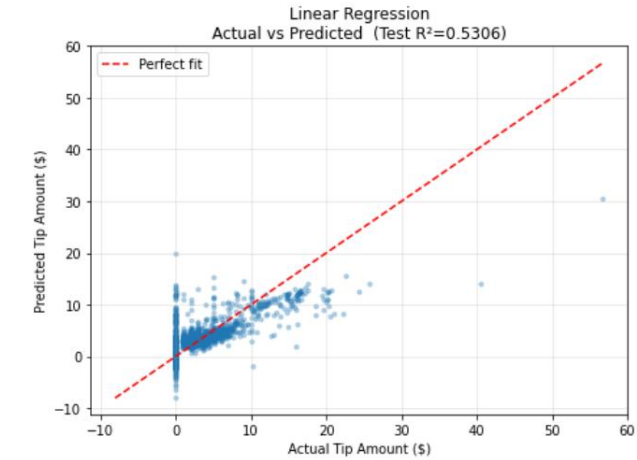


Fig. 4a: Linear Regression: Actual vs. Predicted `tip_amount` (Test $R^2=0.5306$). The vertical cluster near $x=0$ reflects structural zero-tip cash transactions.

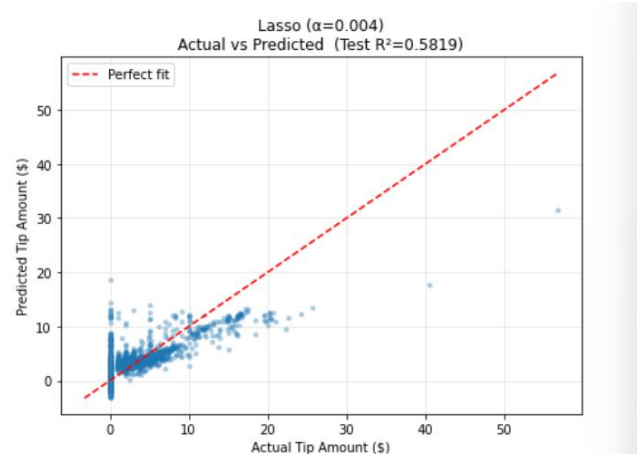


Fig. 4b: Lasso Regression ($\alpha=0.004$): Actual vs. Predicted `tip_amount` (Test $R^2=0.5819$). Tighter clustering around the perfect-fit line compared to OLS indicates improved generalization.

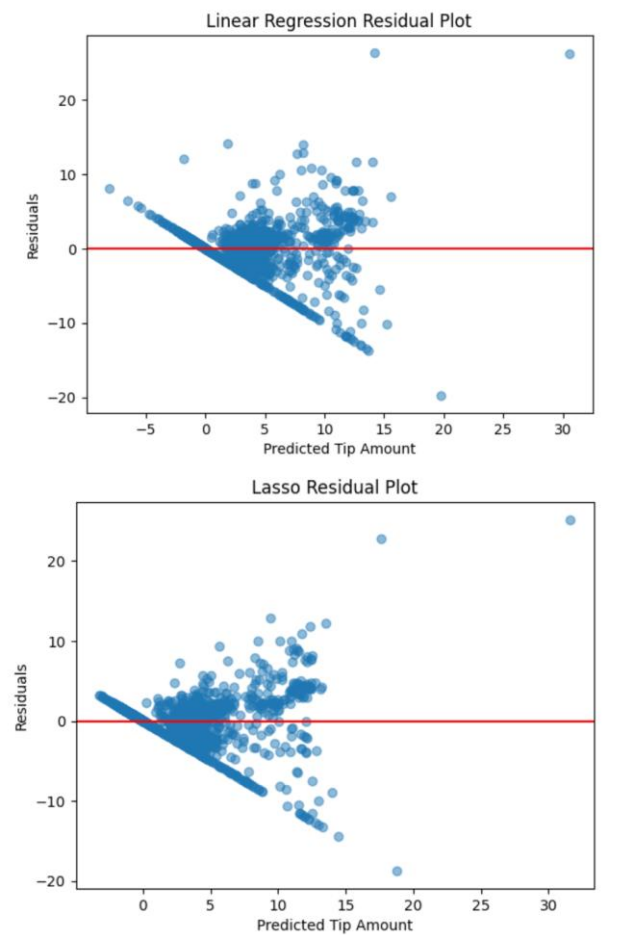


Fig. 5. Residual plots for Linear Regression (top) and Lasso (bottom). The diagonal stripe at negative residuals reflects zero-tip cash trips where the model predicts positive values.

REFERENCES

[1] NYC Taxi and Limousine Commission, "2023 Yellow Taxi Trip Data," NYC Open Data, 2023. [Online]. Available: <https://opendata.cityofnewyork.us/>